EVRYN 20-Paper Benchmark Partition and Strict Zero-Leakage Evaluation Protocol

DEVELOPMENT & TUNING ZONE (Permitted Model Calibration)

Train Split (4 Papers)

35 Gold Relations (19 Dir, 16 Cross)

Papers / Identifiers:

PMC2651894 (p. 5) | PMC2682197 (p. 12)
PMC2714965 (p. 12) | PMC2774163 (p. 13)

- Grammar & trigger discovery
- Syntactic pattern mining
- Rule & invariant calibration
- Strictly isolated from test

Dev Split (8 Papers)

71 Gold Relations (36 Dir, 35 Cross)

Papers / Identifiers:

PMC1804205 (p. 7) | PMC1874608 (p. 12)
PMC1913099 (p. 16) | PMC1974823 (p. 7)
PMC2242835 (p. 10) | PMC2266911 (p. 12)
PMC2358977 (p. 12) | PMC2430206 (p. 10)

- Error census & FP diagnosis
- Threshold tuning (tau = 0.60)
- Criterion 5 verified (0.5350 F1)
- TCSTS reverse-edge gating

[GATE]

ZERO

LEAKAGE

RELEASE

GATE

Code & Rules

Frozen Prior

to Confirmatory

Evaluation

Zero Feedback

Zero Leakage

CONFIRMATORY EVALUATION ZONE (Strictly Held-Out & Blind)

Locked Test (4 Papers)

41 Gold Relations (21 Dir, 20 Cross)

Papers / Identifiers:

PMC2565068 (p. 9) | PMC2581603 (p. 10)
PMC2593050 (p. 5) | PMC2639726 (p. 14)

- Single post-hoc evaluation
- Confirmatory generalization
- Frozen model parameters
- Zero hyperparameter tuning

External Unseen (4 Papers)

36 Gold Relations (20 Dir, 16 Cross)

Papers / Identifiers:

PMC2816692 (p. 10) | PMC2829055 (p. 9)
PMC2858072 (p. 8) | PMC2885601 (p. 6)

- Unseen pathogen genetics
- Authenticated V2 ground truth
- Out-of-domain evaluation
- Zero tuning feedback

Benchmark Ground-Truth Ledger & Annotation Audit Protocol (20 Papers, 183 Gold Causal Relations)

Legacy Benchmark V1 (Unaligned Metadata):

- PMC2816692: Title registered as Anthrax LF while XML described Salmonella SrcA
- Resulted in 54 False Positives on External split; Direct F1 = 0.3269, Cross F1 = 0.2548

Authenticated Benchmark V2 (Audited Ground Truth):

- PMC2816692: Aligned to authentic SrcA/PipB2/SseL mechanisms (15 False Positives)
- 183 Gold Relations (96 Direct, 87 Cross); Ext Direct F1 = 0.4808, Cross F1 = 0.3262